MOMENT OF INERTIA

Rotational Kinetic Energy and Inertia

Just like massive bodies tend to resist changes in their motion (AKA - "Inertia") . Rotating bodies also tend to resist changes in their motion. We call this **ROTATIONAL INERTIA**. We can determine its expression by looking at Kinetic Energy.

$$K = \tfrac{1}{2} m v^2, \quad v_t = r\omega$$

$$K = \tfrac{1}{2} m (r\omega)^2$$

$$K = \tfrac{1}{2} \boxed{mr^2} \omega^2, \quad I = \sum m r^2$$

$$K_{rot} = \tfrac{1}{2} I \omega^2$$

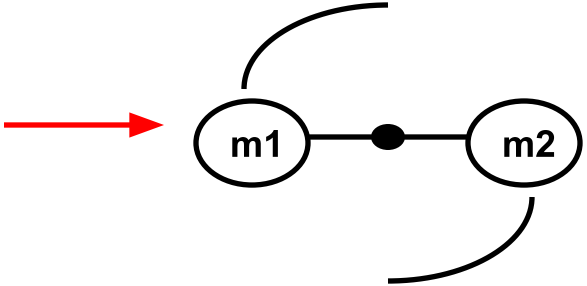
We now have an expression for the rotation of a mass in terms of the radius of rotation.
We call this quantity the **MOMENT OF INERTIA (I)** with units **kg*m^2**

Moment of Inertia, I

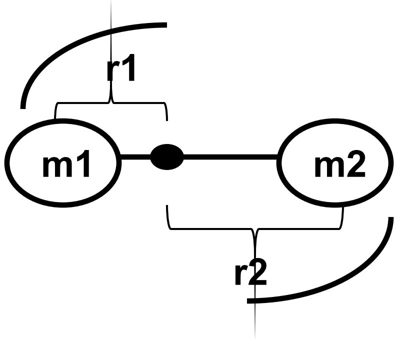
$$I = \sum mr^2$$

Consider 2 masses, m1 & m2, rigidly connected to a bar of negligible mass. The system rotates around its CM.

This is what we would see if m1 = m2.
Suppose m1>m2.



$$v_t = r\omega$$



Since it is a rigid body, the have the SAME angular velocity, ω. The velocity of the center, v_{cm} of mass is zero since it is rotating around it. We soon see that the TANGENTIAL SPEEDS are NOT EQUAL due to different radii.

Moment of Inertia, I

Since both masses are moving they have kinetic energy or rotational kinetic in this case.

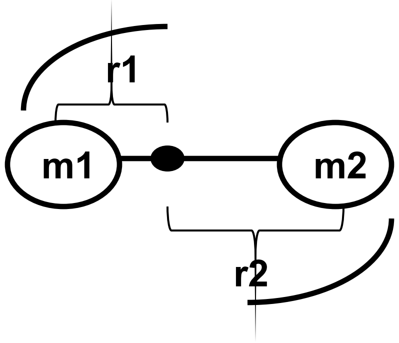
$$K = K_1 + K_2$$

$$K = \tfrac{1}{2} m_1 v_1^2 + \tfrac{1}{2} m_2 v_2^2, \quad v_t = r\omega$$

$$K = \tfrac{1}{2} m_1 (r_1^2 \omega^2) + \tfrac{1}{2} m_2 (r_2^2 \omega^2)$$

$$K = \tfrac{1}{2} (m_1 r_1^2 + m_2 r_2^2) \omega^2$$

$$K = \tfrac{1}{2} \left(\sum_{i=1}^{N} m_i r_i^2\right) \omega^2 \rightarrow K = \tfrac{1}{2} I \omega^2$$

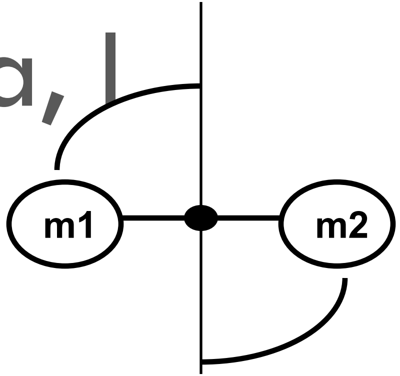


So this example clearly illustrates the idea behind the SUMMATION in the moment of inertia equation.

Example: Moment of Inertia, I

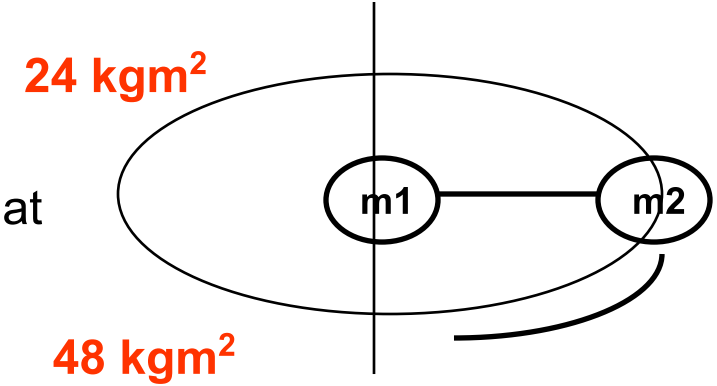
$$I = \sum mr^2$$

Let's use this equation to analyze the motion of a 4-m long bar with negligible mass and two equal masses(3-kg) on the end rotating around a specified axis.



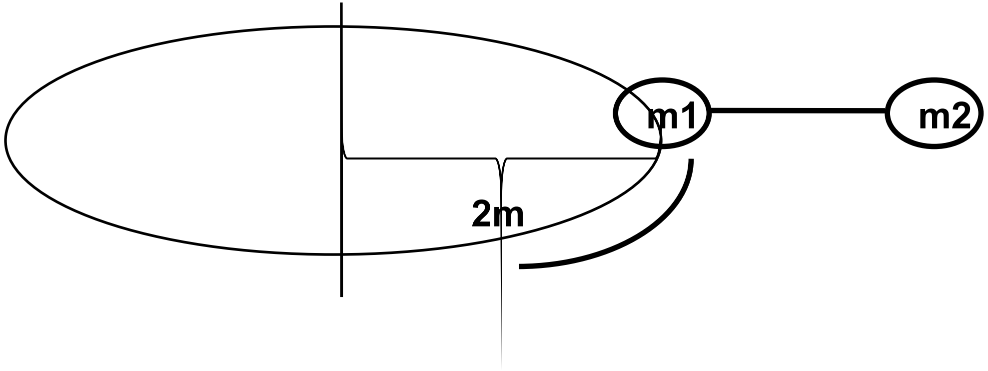
EXAMPLE #1 -The moment of Inertia when they are rotating around the center of their rod.

EXAMPLE #2-The moment if Inertia rotating at one end of the rod



Example cont'

Now let's calculate the moment of Inertia rotating at a point 2 meters from one end of the rod.



120 kgm^2

As you can see, the FARTHER the axis of rotation is from the center of mass, the moment of inertia increases. We need an expression that will help us determine the moment of inertia when this situation arises.

Parallel Axis Theorem

This theorem will allow us to calculate the moment of inertia of any rotating body around any axis, provided we know the moment of inertia about the center of mass.

$$I_p = I_{cm} + Md^2$$

It basically states that the Moment of Inertia **(I_p)** around any axis "P" is equal to the known moment of inertia **(I_{cm})** about some center of mass plus **M** (the total mass of the system) times the square of "**d**" (the distance between the two parallel axes)

Using the prior example let's use the parallel axis theorem to calculate the moment of inertia when it is rotating around one end and 2m from a fixed axis.

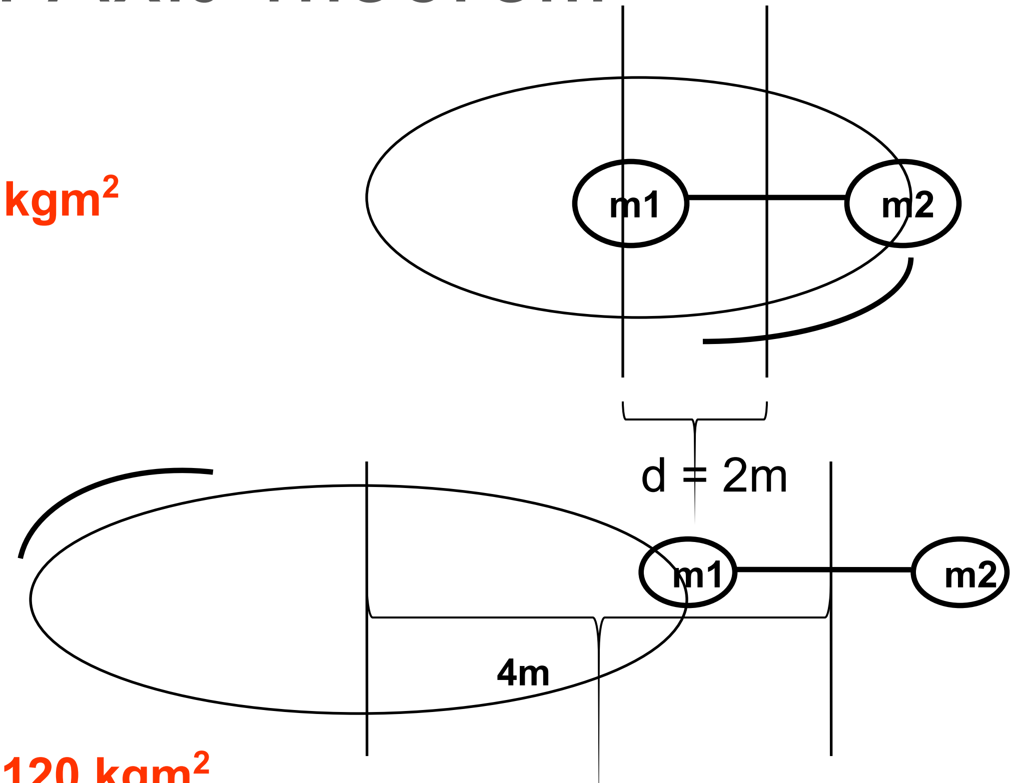
Exam – Parallel Axis Theorem

$$I_p = I_{cm} + Md^2$$

$$I_p = (24) + (6)(2)^2 =$$ **48 kgm²**

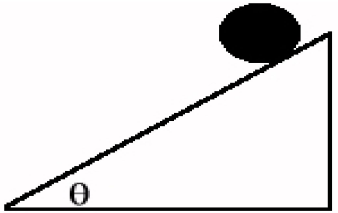
$$I_p = I_{cm} + Md^2$$

$$I_p = (24) + (6)(4)^2 =$$ **120 kgm²**



(not drawn to scale)

Example



A very common problem is to find the velocity of a ball rolling down an inclined plane. It is important to realize that you cannot work out this problem the way you used to. In the past, everything was SLIDING. Now the object is rolling and thus has MORE energy than normal. So let's assume the ball is like a thin spherical shell and was released from a position 5 m above the ground. Calculate the velocity at the bottom of the incline.

$$E_{before} = E_{after}$$
$$U_g = K_T + K_R$$
$$mgh = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$$
$$v = R\omega \qquad I_{sphere@cm} = \tfrac{2}{3}mR^2$$
$$mgh = \tfrac{1}{2}mv^2 + \tfrac{1}{2}(\tfrac{2}{3}mR^2)(\frac{v^2}{R^2})$$
$$gh = \tfrac{1}{2}v^2 + \tfrac{1}{3}v^2$$
$$v = \sqrt{\tfrac{6}{5}gh} = \sqrt{\tfrac{6}{5}(9.8)(5)} =$$ **7.67 m/s**

$$E_{before} = E_{after}$$
$$U_g = K_T$$
$$mgh = \tfrac{1}{2}mv^2$$
$$mgh = \tfrac{1}{2}mv^2$$
$$gh = \tfrac{1}{2}v^2$$
$$v = \sqrt{2gh} = \sqrt{2(9.8)(5)} = 9.90\ m/s$$

If you HAD NOT included the rotational kinetic energy, you see the answer is very much different.